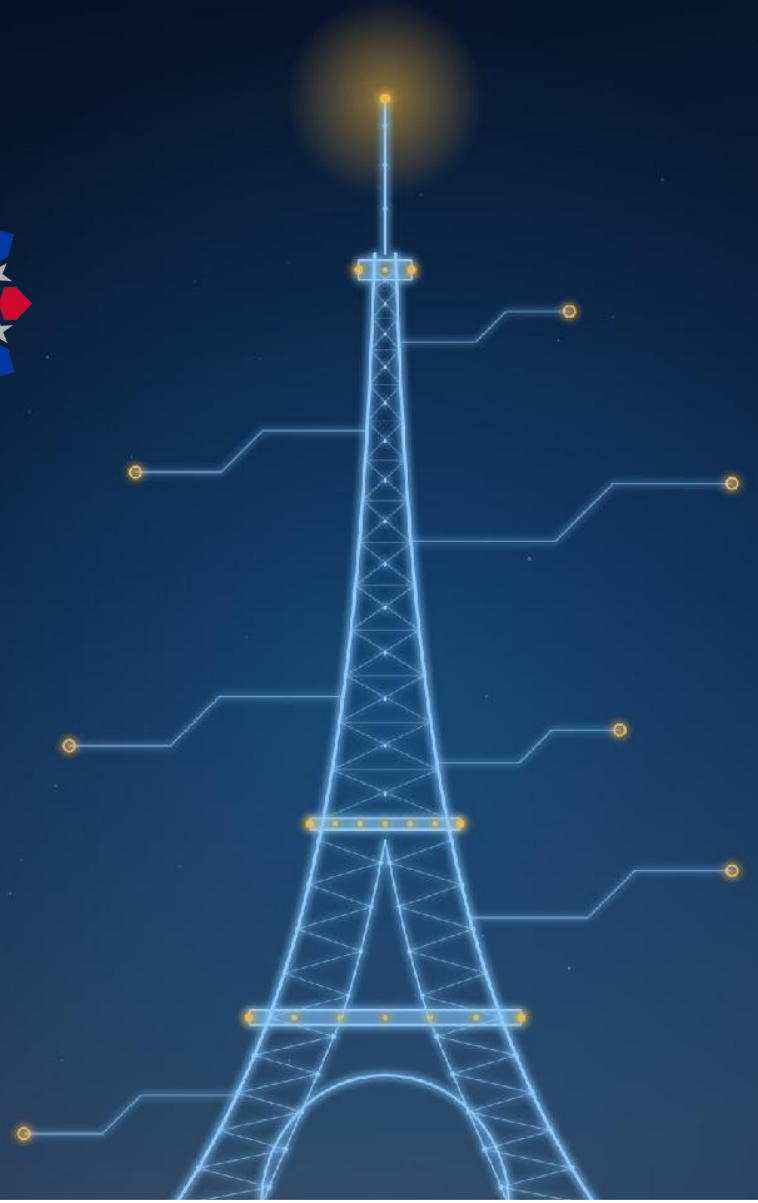
ETECOM 2026 · PARIS

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A Comparative Study of Fine-Tuned MARBERTv2 for Arabic Dialect Identification on the IADD Benchmark

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Outline

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02 **Dataset & Methodology**

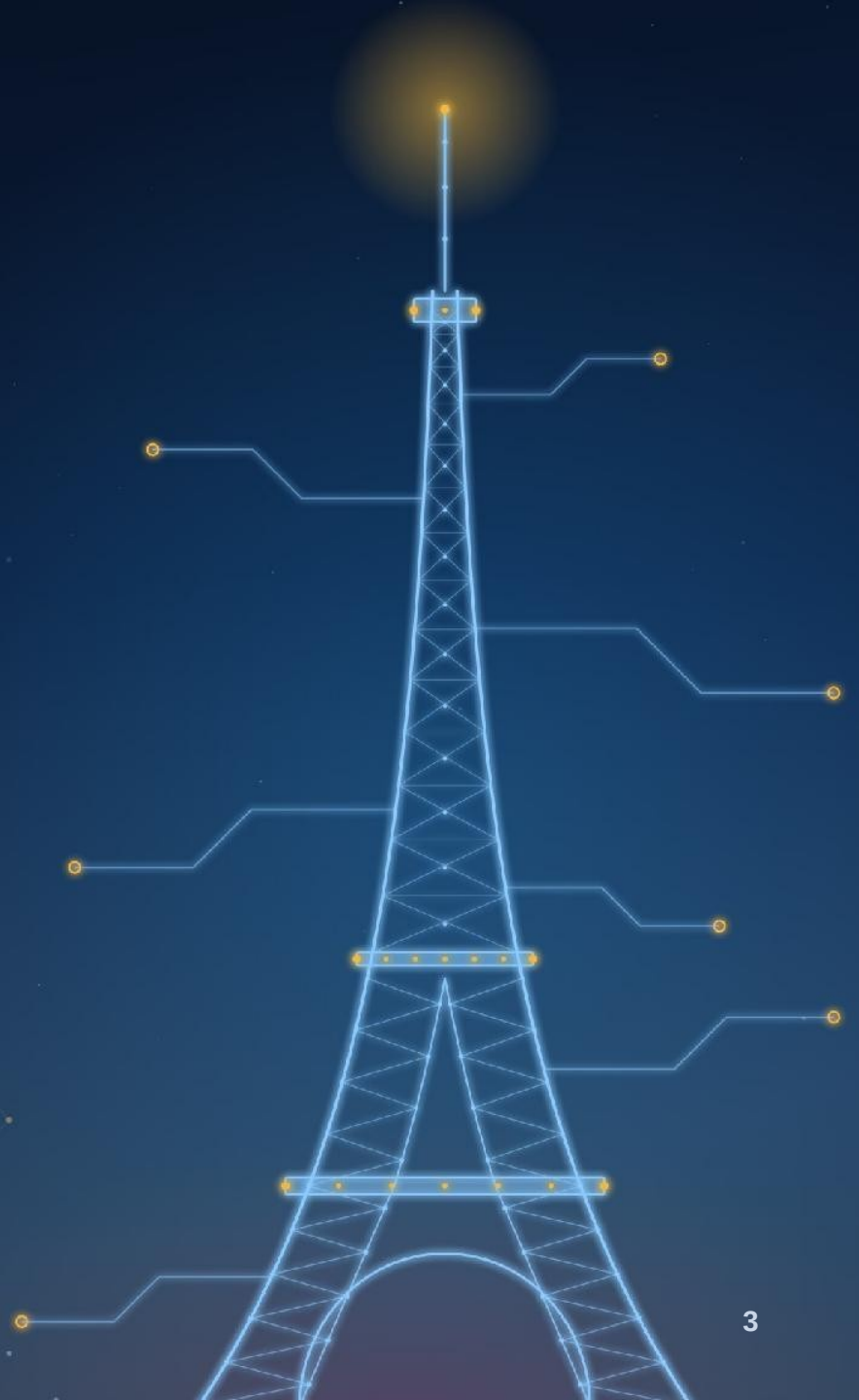
03 **Results**

04 **Discussion & Limitations**

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01

Introduction & Motivation



Motivation: Same Questions, Different Dialects

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- 500M+ speakers, Modern Standard Arabic (MSA) rarely spoken
- NLP systems trained on MSA struggle with dialects
- Arabic dialects defer in vocabulary, spelling, grammar, word choice:

TABLE I · Dialectal Differences

Dialect	Arabic	Transliteration	Meaning	Dialect
MSA	ماذا تريد	*mādhā turīd*	What do you want?	MSA
Egyptian	عايز إيه	*ʿāyez eh*	What do you want?	Egyptian
Gulf	وش تبي	*wesh tebi*	What do you want?	Gulf
Levantine	شو بدك	*shu biddak*	What do you want?	Levantine

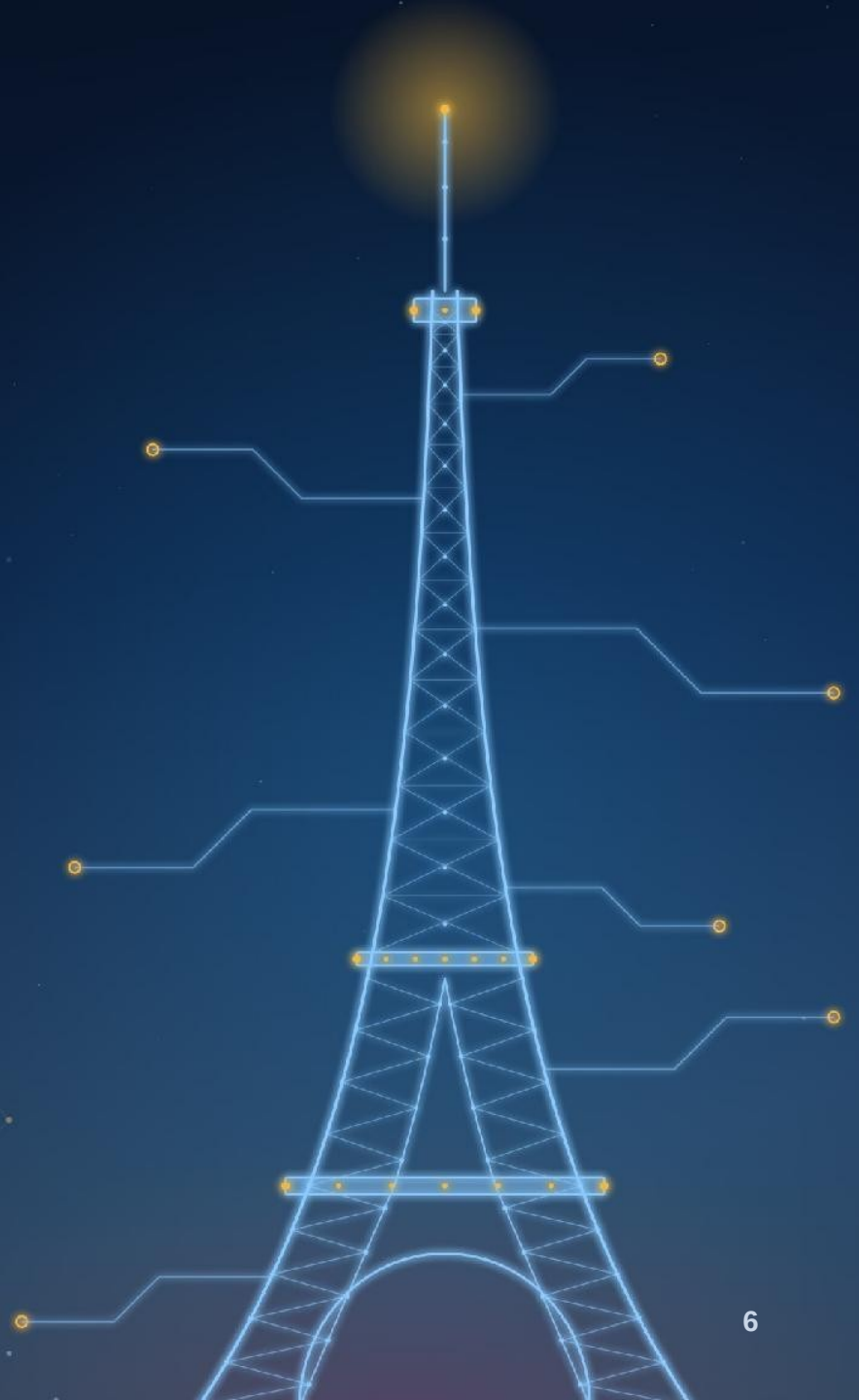
Gap and Contributions

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- **What's known:**
  - Early work mostly uses TF-IDF. Naïve Bayes was best on NADI 2021 [4]; word + character features yields better results
  - Transformers reach higher accuracy: 93.06% on IADD [10], hybrid models reach 96.54% [9]
- **Gap:**
  - Accuracy vs cost trade-off is never mentioned directly
  - Transformers are more accurate, but is the gain worth the cost?

02

Dataset and Methodology



Dataset

- IADD, merges 5 corpora: AOC, DART, SHAMI, TSAC, PADIC
- 4 dialects out of 5 were kept
- **Cleaning**
  - Arabic Normalization
  - Non Arabic characters removed
  - Empty samples dropped
  - Duplicates removed

- **Imbalance:** Egyptian had 4,555 while Levantine had 84,202
  - Therefore, training set is balanced at 3,280 per dialect

TABLE II · Samples Per Dialect for each Split

Dialect	Train (balanced)	Train (capped)	Validation	Test
Egyptian	3,280	3,280	364	911
Gulf	3,280	5,341	364	911
Levantine	3,280	12,000	364	911
Maghrebi	3,280	12,000	364	911

Methodology

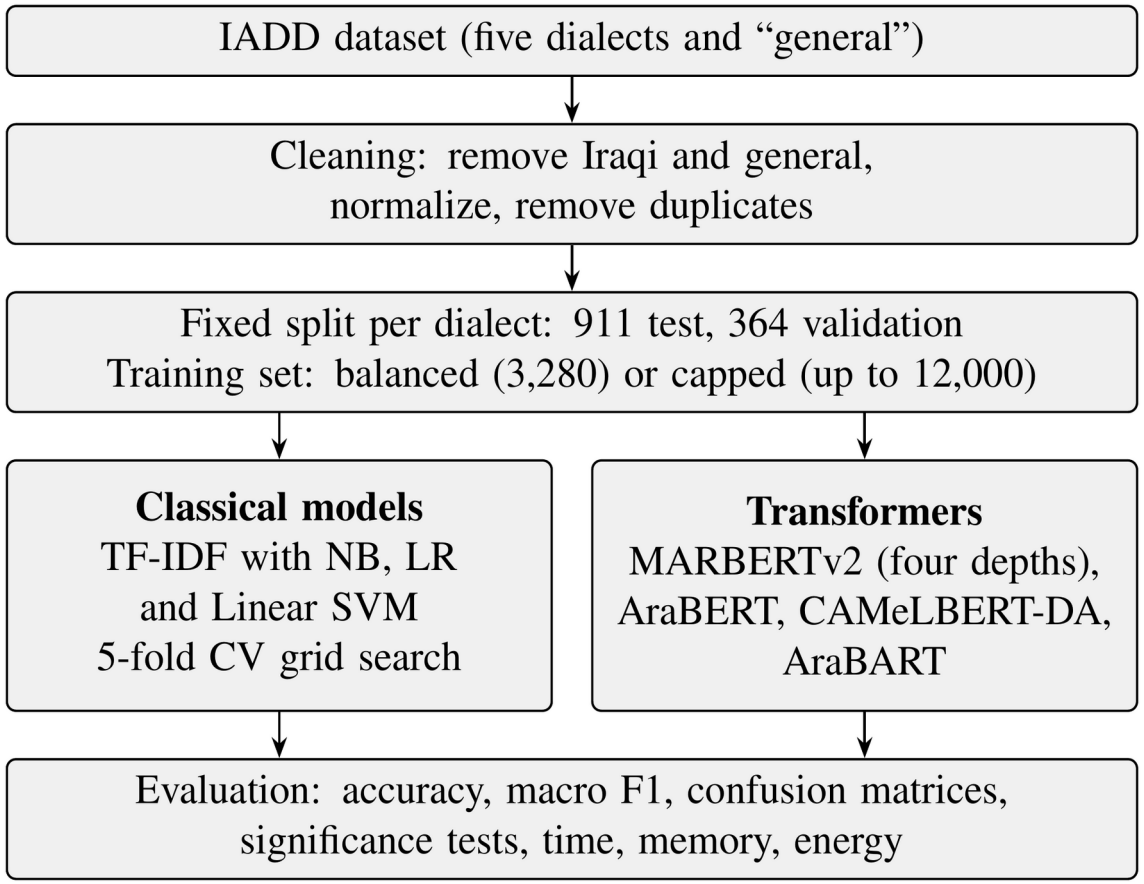
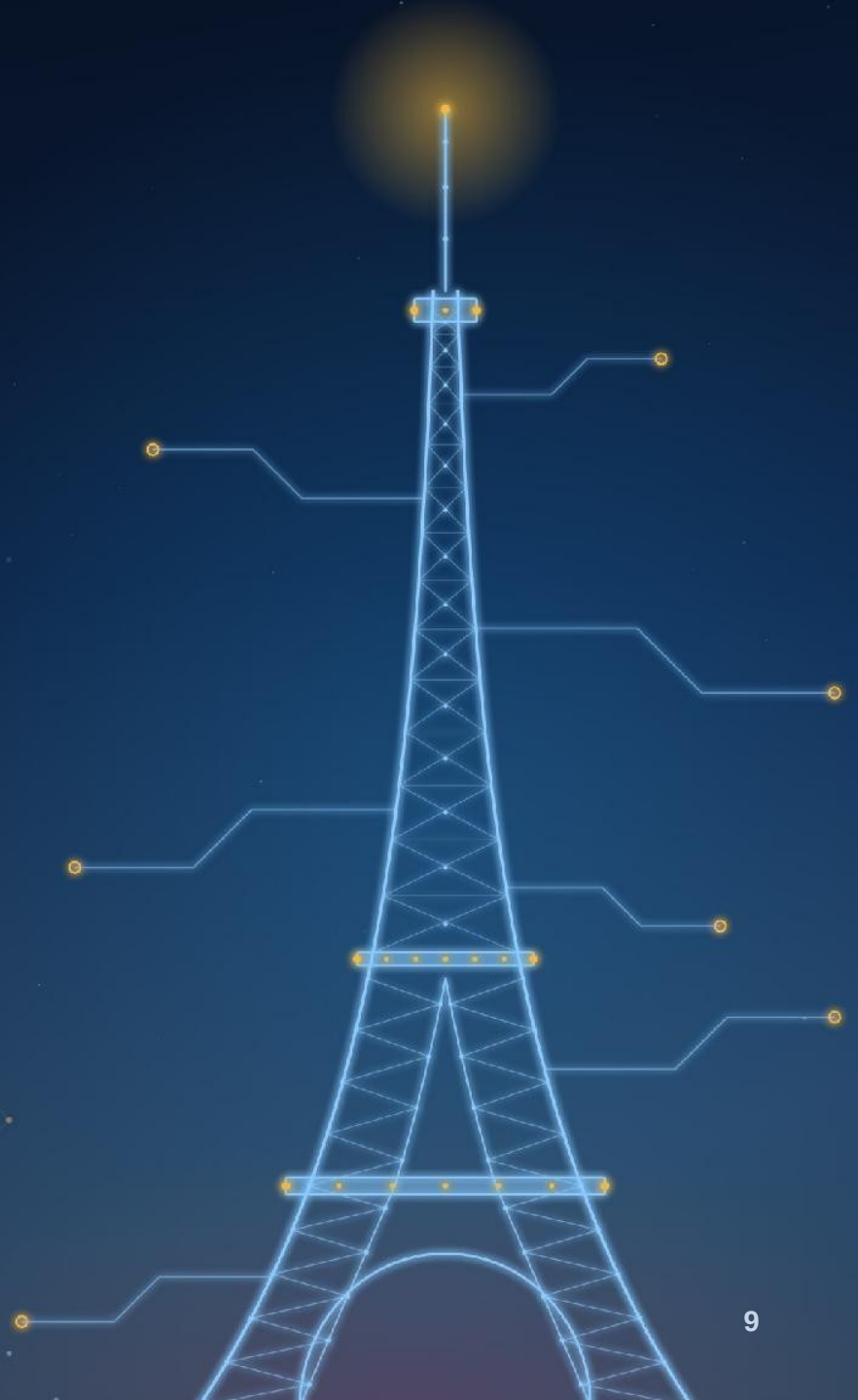


Fig.1: Overview of the experimental pipeline

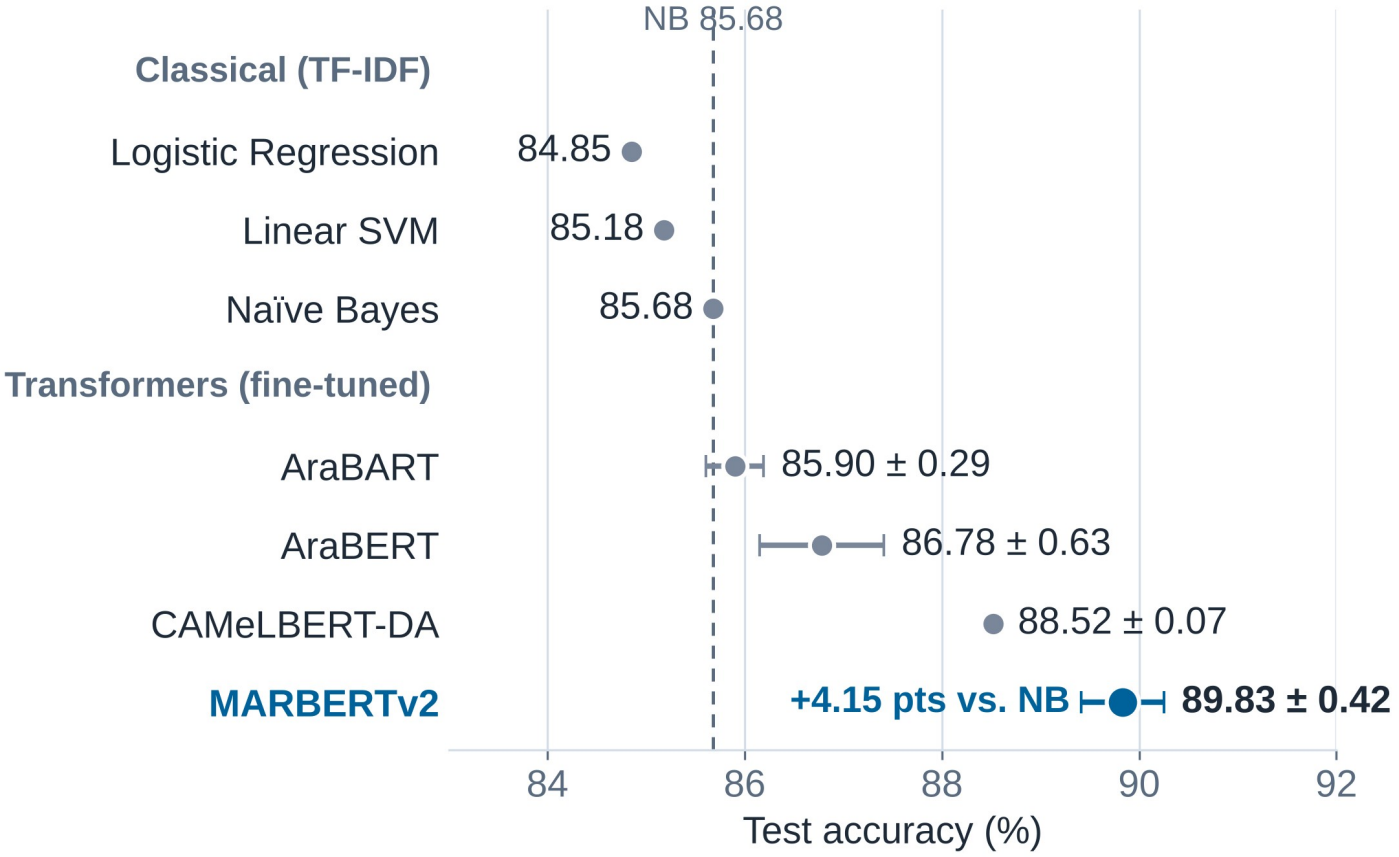
- **MARBERTv2:** trained on approx. 1B tweets, fine tuned for 4 epochs with learning rate chosen at validation
- **Fair comparison:** same machine, checkpoint picked on validation, test set used once, 5 seeds for MARBERT, 3 for other transformers
- **Classical models:** turned with CV 5-fold grid search on word and character TF-IDF

03

Results



Overall Accuracy



- **+4.15 points** over Naïve Bayes, significant in all 5 seeds ($p < 0.001$)
- Classical models land within 0.83 points, so the algorithm barley matters
- Dialect-pretrained transformers (MARBERTv2, CAMeLBERT-DA) lead the other two

Fig. 2. Test accuracy of all seven models. (transformers: mean ± std over seeds)

Accuracy vs. Cost

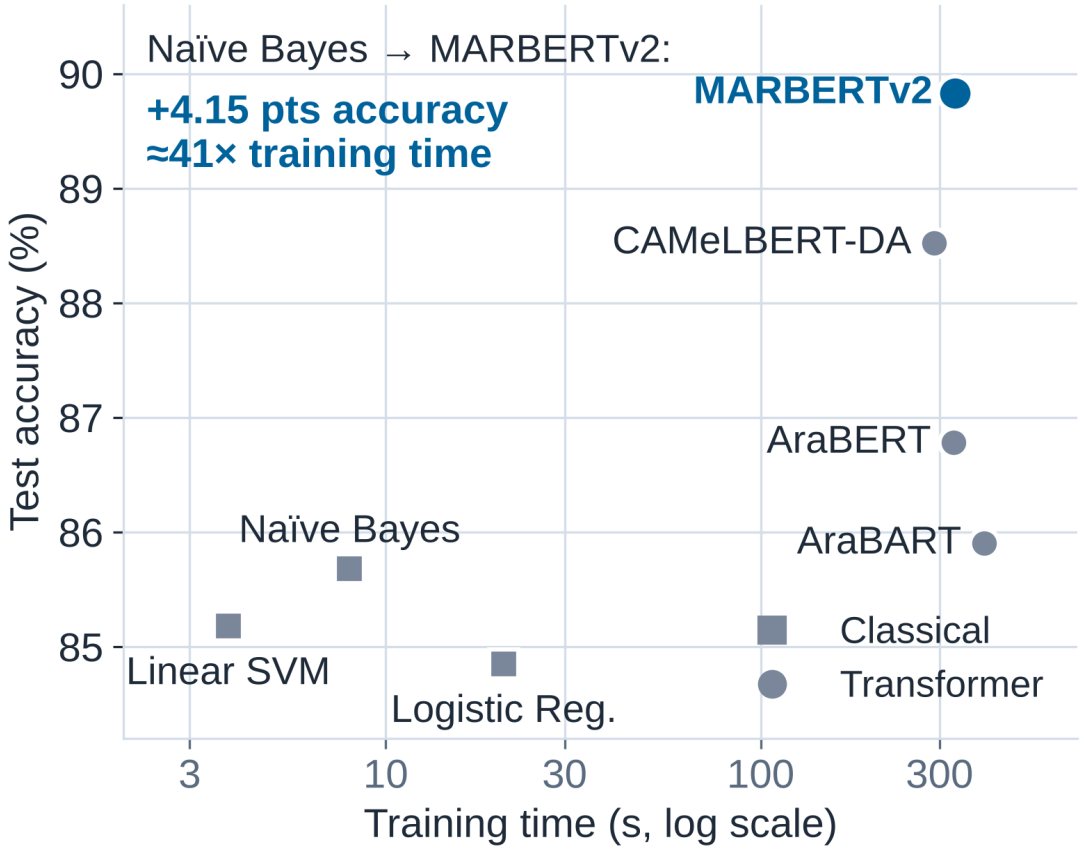


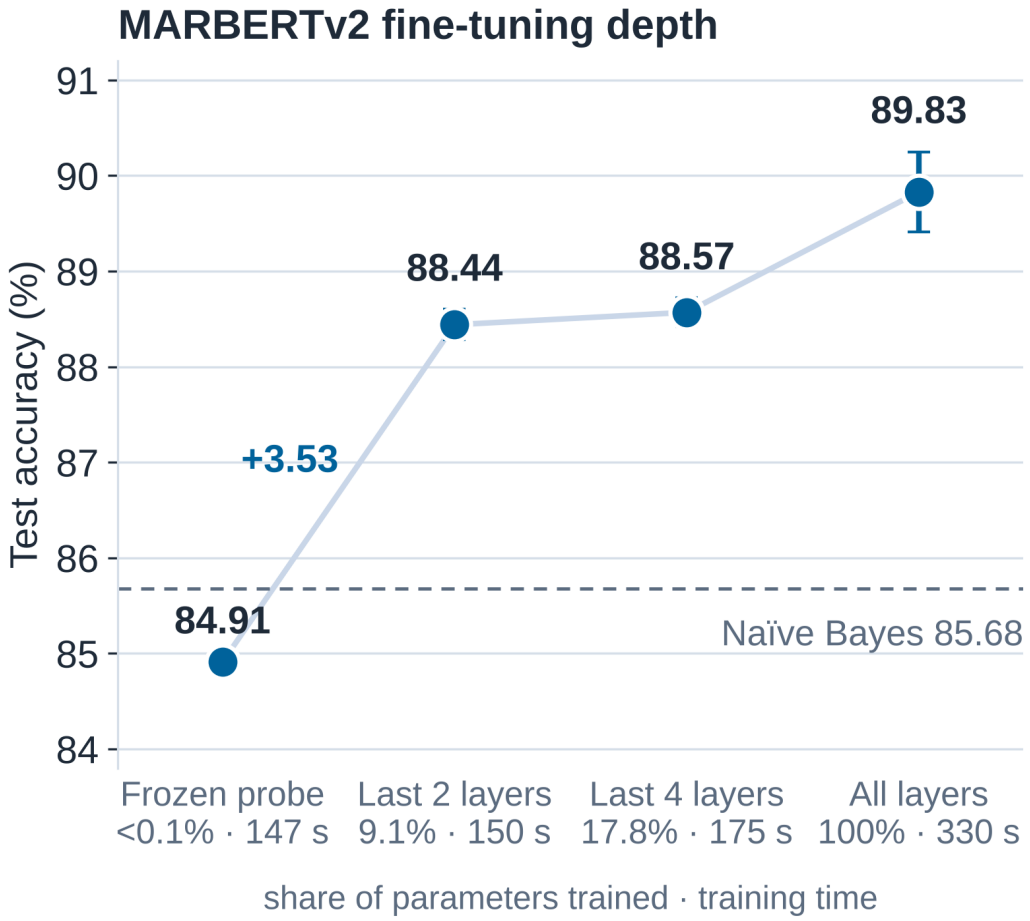
Fig. 3. Test accuracy vs. Training time (log scale), same machine

TABLE III · Computational cost of Naïve Bayes vs. MARBERTv2 on the same machine

	Naïve Bayes	MARBERTv2
Training	8.0 s	330 s (≈41x)
Inference (3,644 texts)	1.9 s	3.9 s (1.1 ms/text)
Energy per training run	0.12 Wh	7.24 Wh (≈60x)

- **The main cost is training.** Inference is only 2-6x slower
- Real-time, high-volume use favors classical models; accuracy-first use favors MARBERTv2

Fine-tuning depth and focal loss



- **Frozen probe (84.91%) is below Naïve Bayes, so the gain isn't in the embeddings alone**
- **Last 2 layers** recover about ⅔ of the gain in under half the training time.
- **Focal loss on about 2.5x more data** didn’t help (89.05% vs. 89.83%, p = 0.11)

Fig. 4. MARBERTv2 test accuracy at four fine-tuning depths (dashed line marks Naïve Bayes (85.68%))

Per-dialect results

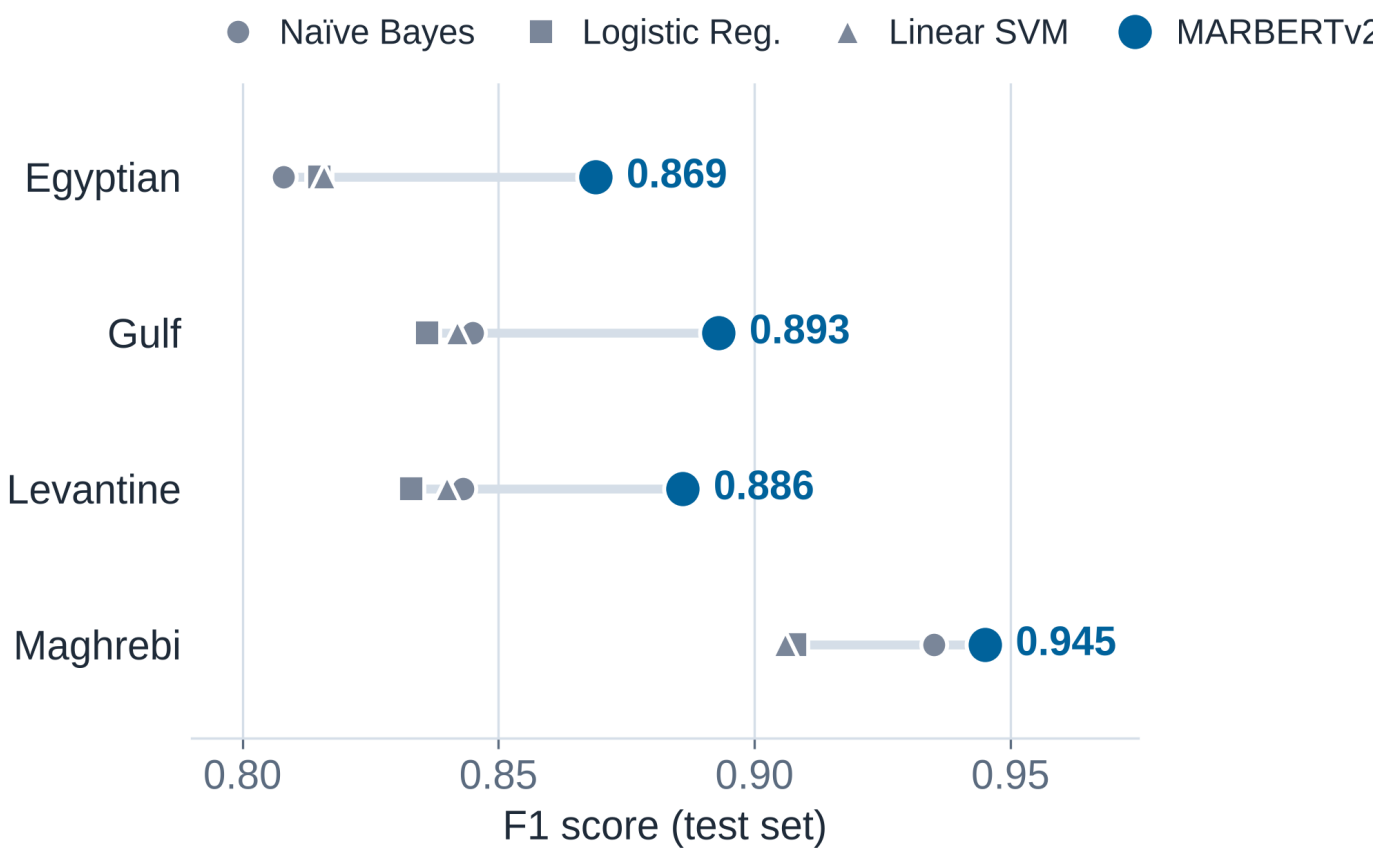


Fig. 5. Per-dialect F1 of the three classical models & MARBERTv2

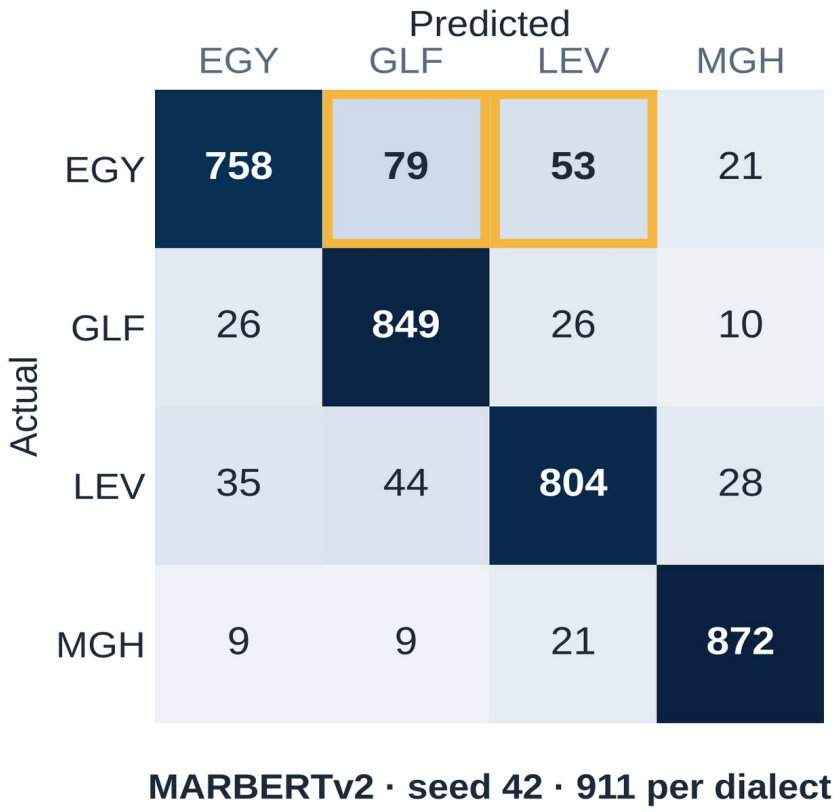
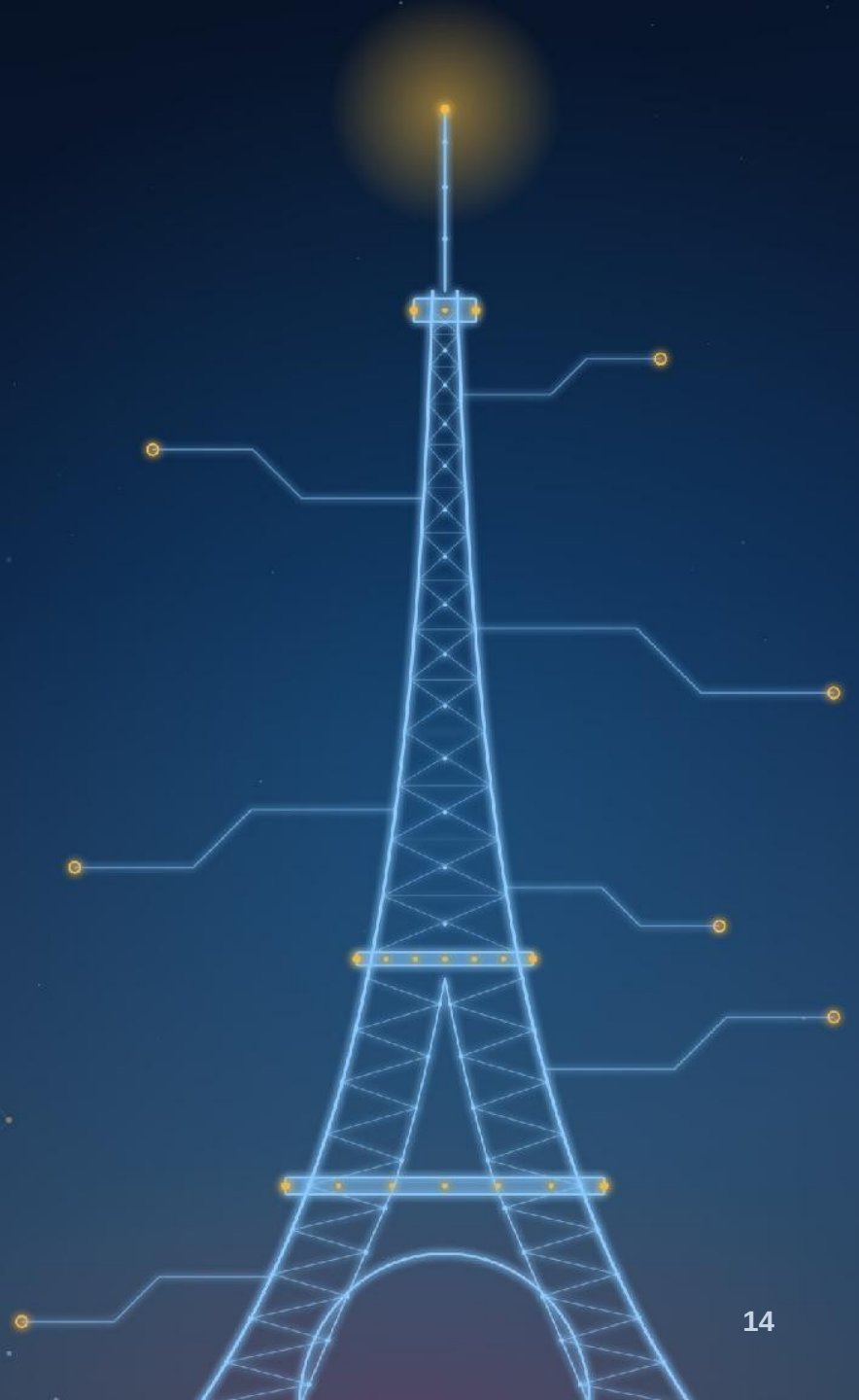


Fig. 6. Confusion matrix of MARBERTv2 on the test set

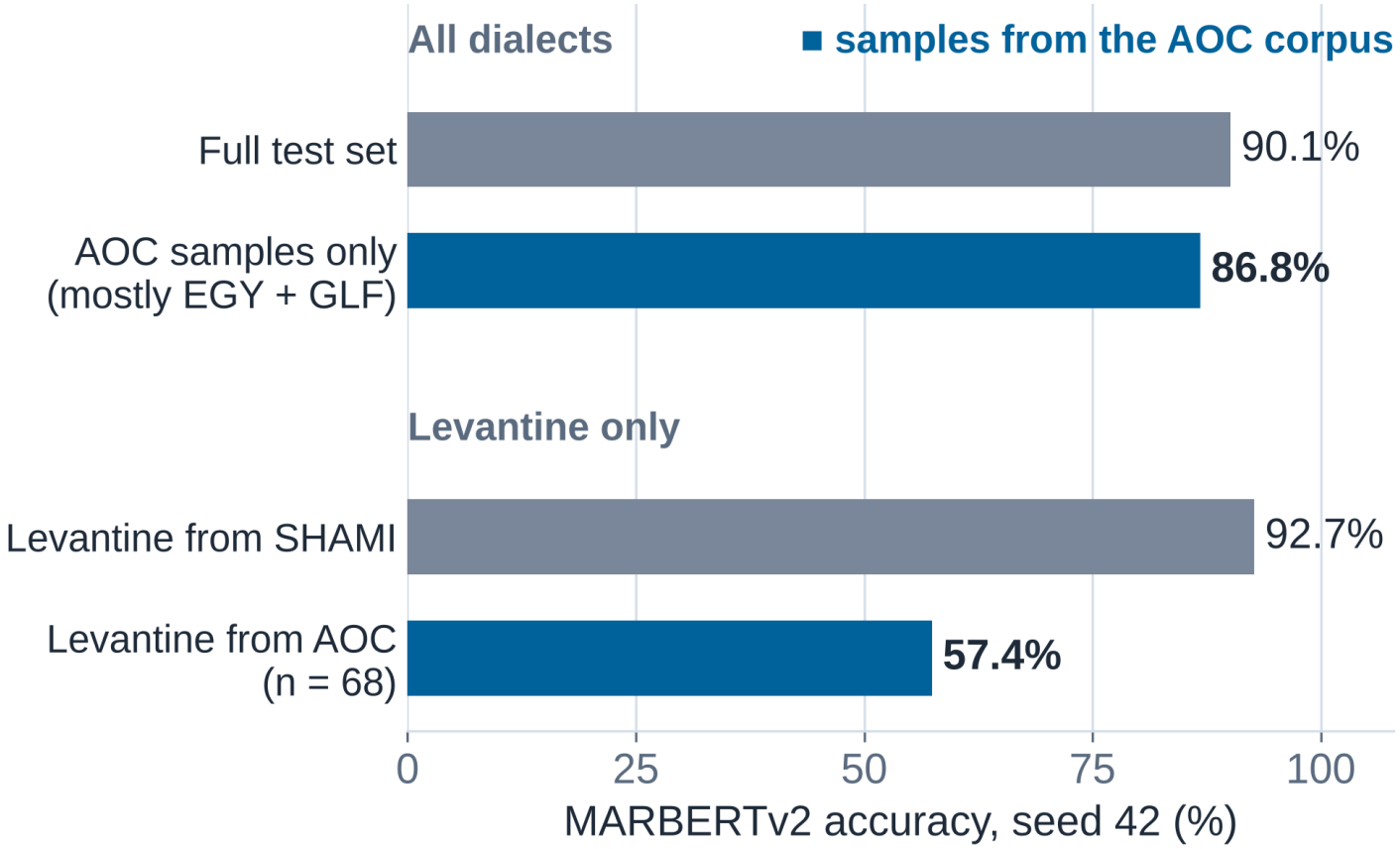
- Maghrebi easiest, Egyptian hardest, in every model
- Egyptian is mostly confused with Gulf (79) and Levantine (53)

04

Discussions & Limitations



Dialect or dataset?



- **The same ranking appears in every model, so it isn't caused by any one model**
- **The dialect explanation:** vocabulary overlap (Jaccard on top-300 TF-IDF terms) is highest for Egyptian-Gulf (0.313) and lowest between Maghrebi and every other dialect.
- **The dataset explanation:** dialect and source are linked in IADD. Egyptian and Gulf come mostly from AOC, and Maghrebi almost entirely from PADIC and TSAC

Fig. 7. MARBERTv2 test accuracy (seed 42) on subsets grouped by source corpus. Samples from AOC are harder, most sharply for Levantine

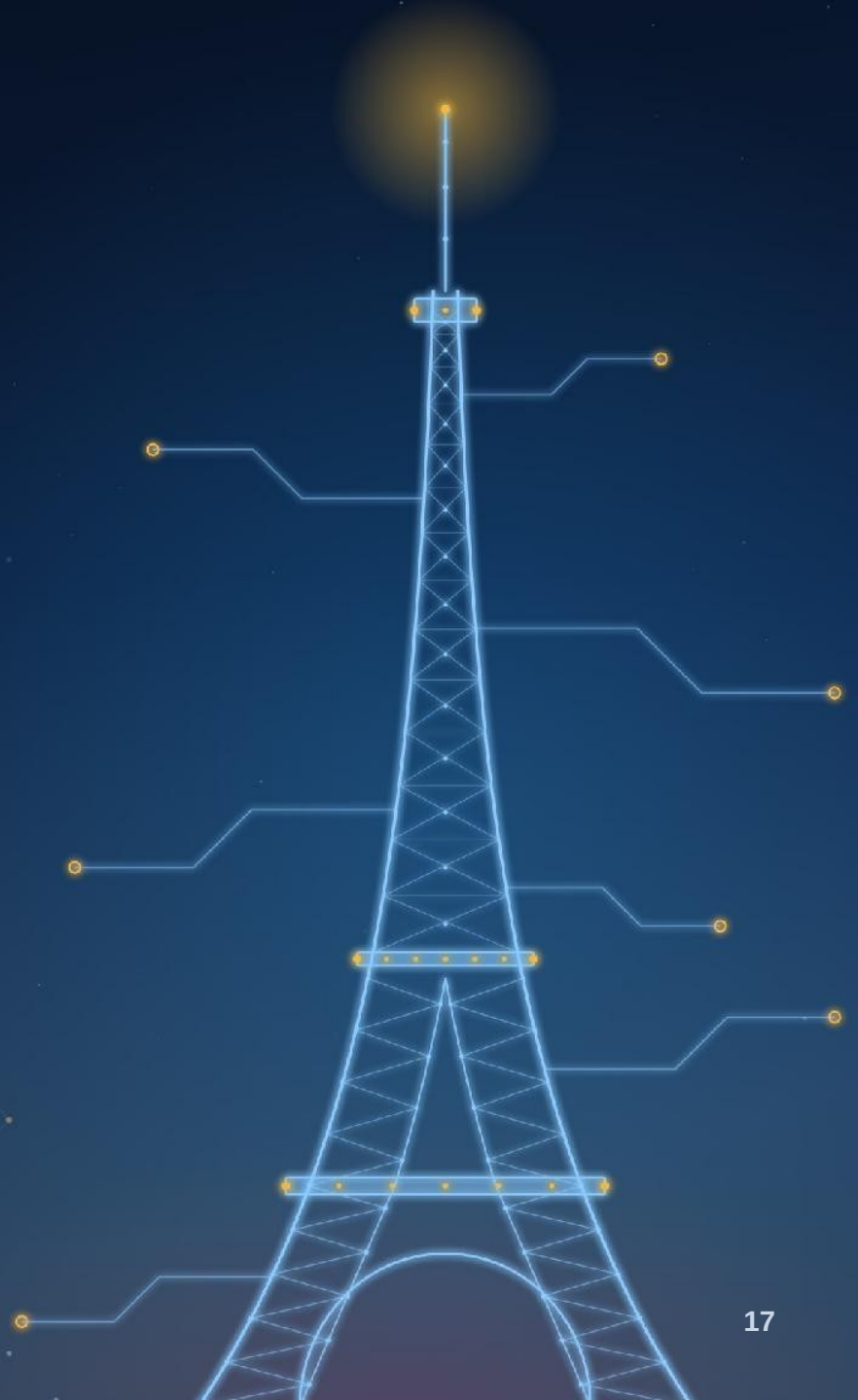
Limitations

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- **Unequal tuning:** The learning rate was seached only for MARBERTv2. The other transformers resued it and ran 3 seeds instead of 5
- **Source confound:** Dialect source corpus are linked in IADD
- **Focal-loss run:** It changed the loss and the training data size together, so the effect of each can't be separated
- **No Iraqi: I**t was excluded with only 216 samples, so the results don't extend to Iraqi Arabic.
- **Curated data:** Results may not hold on noisy, code-switched text, or on sentences that fit more than one dialect [17]

05

Conclusion and Future Work



Conclusion & Future Work

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Conclusions

- MARBERTv2 reaches 89.93% vs NB's 85.68%
- Gain costs more training time and energy
- Fine-tuning last 2 layers gives best value
- Top two models are the ones pretrained on dialects

Future Work

- Reintroduce Iraqi with more samples
- Collect all dialects from same kinds of sources to separate dialect effects from source bias
- Test on noisy real-world text

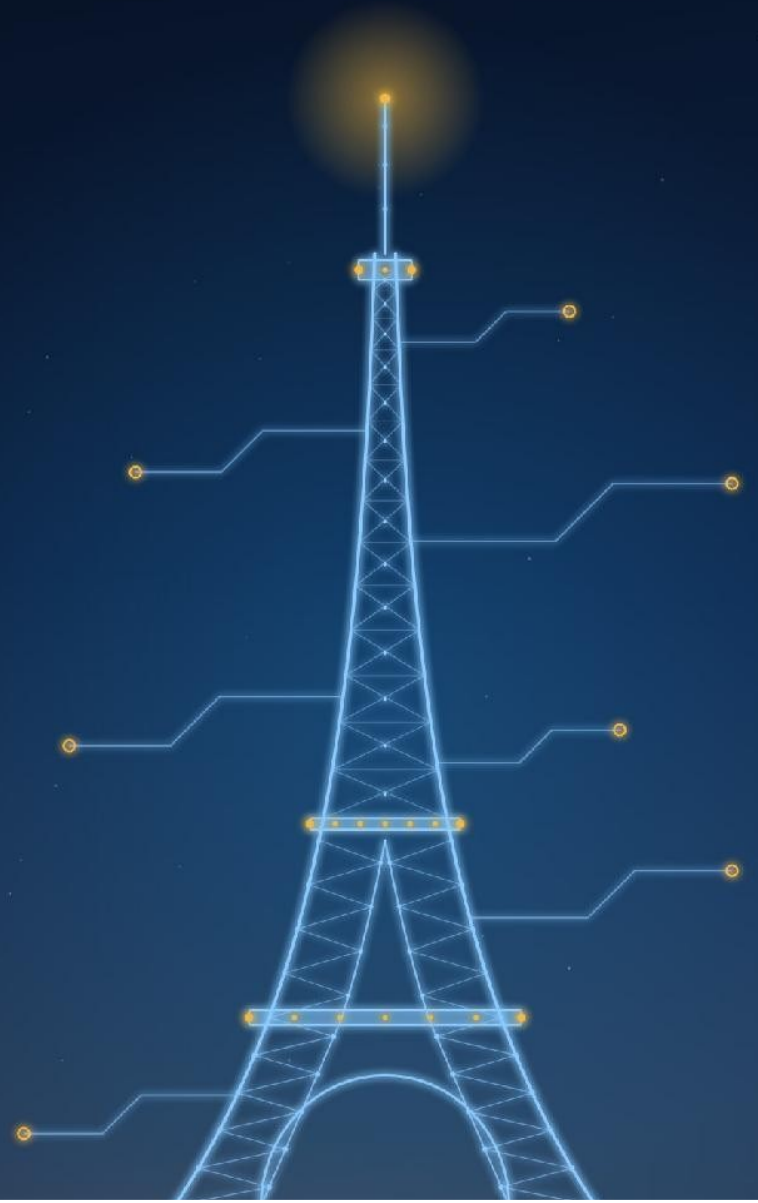
Thank you

Questions & discussion

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Backup

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BACKUP A · Accuracy and cost of all models on the test set (paper, Table II)

Model	Acc. (%)	Macro F1	Train (s)	Infer. (s)	Samples/s	GPU (GB)	Size (MB)	Energy (Wh)
Classical: TF-IDF, word + char n-grams								
Naïve Bayes	85.68	0.858	8.01	1.86	1,962	–	11.4	0.12
Logistic Regression	84.85	0.848	20.66	0.65	5,640	–	7.9	0.25
Linear SVM	85.18	0.851	3.80	0.63	5,747	–	7.9	0.05
Transformers: full fine-tuning (mean ± std over seeds)								
MARBERTv2 (5 seeds)	**89.83 ± 0.42**	**0.898**	**330**	**3.89**	**936**	**2.83**	**621**	**7.24**
AraBERT (3 seeds)	86.78 ± 0.63	0.867	327	4.53	804	2.53	516	7.15
CAMeLBERT-DA (3 seeds)	88.52 ± 0.07	0.885	290	4.48	814	2.24	416	6.57
AraBART (3 seeds)	85.90 ± 0.29	0.859	394	5.25	694	2.88	533	8.52
MARBERTv2 variants								
Frozen probe (<0.1% trained)	84.91	0.848	147	4.21	865	0.80	621	2.21
Last 2 layers (9.1%)	88.44 ± 0.16	0.884	150	3.89	937	1.06	621	3.16
Last 4 layers (17.8%)	88.57 ± 0.15	0.886	175	3.87	942	1.19	621	4.02
Capped set + focal loss	89.05 ± 0.50	0.890	754	3.97	918	2.83	621	16.18

BACKUP B · Training settings (paper, Table I and Sec. III-C)

Setting	Value
MARBERTv2 fine-tuning	
Epochs · optimizer	4 · AdamW
Learning rate	**5×10^{-5} (from 1, 2, 3, 5×10^{-5})**
Batch (train / infer.)	16 / 64
Warm-up · decay	10% linear · 0.01
Precision · length	fp16 · 128 tokens
Seeds	42–46
Selection	best validation macro F1
Classical: 5-fold CV grid (macro F1)	
NB α	0.01, 0.1, 0.5, 1 → 0.1
LR C	0.1, 1, 10 → 1
SVM C	0.1, 1, 10 → 0.1
Features	word 50k + char 2–5 100k

Times and throughput from the seed-42 run; accuracy, F1, GPU memory and energy averaged over seeds. All runs on one Colab machine (Xeon 2.0 GHz, 2 cores, 12.7 GB RAM, Tesla T4 15 GB).

BACKUP C · Significance and robustness

95% CI, seed 42	MARBERTv2 89.10–91.05% vs. NB 84.60–86.77%
McNemar vs. NB	$p < 0.001$ in 5 of 5 seeds; vs. CAMeLBERT-DA: $p < 0.05$ in 2 of 3
3 random splits	MARBERTv2 88.93 ± 1.03% vs. NB 85.72 ± 0.56% (+3.2 pts)